

ADVANCED ROUTING ARCHITECT: EXPERT CURRICULUM v2.0

MODULE 1: OSPF SCALABILITY & FAST CONVERGENCE

In Enterprise networks, OSPF convergence time is critical. Standard OSPF timers are too slow for real-time apps. We use LSA Throttling and SPF Tuning to achieve sub-second convergence.

LSA Type 3 (Summary) and Type 5 (External) must be carefully filtered to keep the LSDB small.

```
router ospf 1
router-id 1.1.1.1
timers throttle spf 50 200 5000
timers throttle lsa all 0 500 5000
area 10 stub no-summary
interface Gi0/1
ip ospf network point-to-point
ip ospf bfd
```

MODULE 2: BGP POLICY CONTROL & TRAFFIC ENGINEERING

BGP is a policy-driven protocol. To control INBOUND traffic, we use AS-Path Prepending or MED.

To control OUTBOUND traffic, we use Local Preference.

Expert designs use BGP Communities to signal policies to ISPs without manual intervention.

```
router bgp 65001
neighbor 10.10.10.2 remote-as 65100
address-family ipv4
neighbor 10.10.10.2 route-map PREPEND_OUT out
!
route-map PREPEND_OUT permit 10
set as-path prepend 65001 65001 65001
!
# Verify: show ip bgp neighbors 10.10.10.2 advertised-routes
```

MODULE 3: VRF-LITE & ADVANCED MULTI-TENANCY

VRF-Lite allows multiple routing tables on a single router.

The modern 'vrf definition' command supports both IPv4 and IPv6 families.

Route-Targets (RT) and Route-Distinguishers (RD) are the pillars of this architecture.

```
vrf definition TENANT_A
rd 100:1
address-family ipv4
route-target export 100:1
route-target import 100:1
!
interface Gi0/1.100
vrf forwarding TENANT_A
ip address 10.1.100.1 255.255.255.0
```

MODULE 4: ADVANCED REDISTRIBUTION & LOOP PREVENTION

Multipoint redistribution (OSPF to BGP and vice-versa) is dangerous.

We use Route-Tagging to identify where a route originated and prevent it from 're-entering' the original protocol.

```
route-map OSPF_TO_BGP deny 10
match tag 999
```

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```
route-map OSPF_TO_BGP permit 20
  set tag 111
!
router bgp 65001
 redistribute ospf 1 route-map OSPF_TO_BGP
```

MODULE 5: QoS FOR REAL-TIME APPLICATIONS

QoS ensures Voice (DSCP EF) and Video (DSCP AF41) get priority.
We use MQC (Modular QoS CLI) to Classify, Mark, and Policy/Shape traffic.

```
class-map match-any VOICE_CLASS
  match dscp ef
!
policy-map ENTERPRISE_QOS
  class VOICE_CLASS
    priority percent 30
  class class-default
    fair-queue
!
interface Gi0/1
  service-policy output ENTERPRISE_QOS
```